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(12) **United States Design Patent**
Hikima et al.

(10) **Patent No.:** **US D1,091,652 S**
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(54) **REFRIGERATOR**

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(**) Term: **15 Years**

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(30) **Foreign Application Priority Data**

Oct. 13, 2023 (JP) 2023-021198 D

(51) **LOC (15) Cl.** **15-07**

(52) **U.S. Cl.**
USPC **D15/86**

(58) **Field of Classification Search**
USPC D15/79–91
(Continued)

(56) **References Cited**

U.S. PATENT DOCUMENTS

D700,222 S * 2/2014 Sin D15/89
D729,287 S * 5/2015 Lee D15/89
(Continued)

Primary Examiner — Khawaja Anwar

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(57) **CLAIM**

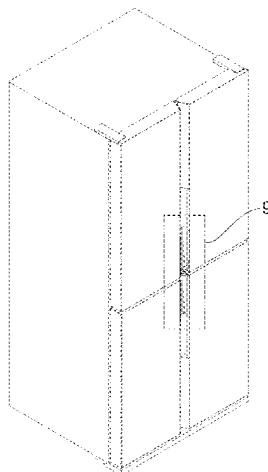
The ornamental design for a refrigerator as shown and described.

DESCRIPTION

FIG. 1 is a front, top, left side perspective view of a refrigerator showing our new design;

FIG. 2 is a front, bottom, right side perspective view thereof; FIG. 3 is a front view thereof; FIG. 4 is a rear view thereof; FIG. 5 is a top view thereof; FIG. 6 is a bottom view thereof; FIG. 7 is a right side view thereof; FIG. 8 is a left side view thereof; FIG. 9 is an enlarged view of portion within the dot-dot-dash broken line box 9 shown in FIG. 1; FIG. 10 is an enlarged view of portion within the dot-dot-dash broken line box 10 shown in FIG. 2; FIG. 11 is an enlarged view of portion within the dot-dot-dash broken line box 11 shown in FIG. 3; FIG. 12 is an enlarged view of portion within the dot-dot-dash broken line box 12 shown in FIG. 11; FIG. 13 is an enlarged view of portion within the dot-dot-dash broken line box 13 shown in FIG. 3; FIG. 14 is an enlarged view of portion within the dot-dot-dash broken line box 14 shown in FIG. 11; FIG. 15 is an enlarged cross-sectional view along the line 15-15 in FIG. 5, in the portion within the dot-dot-dash broken line box 15 in the FIG. 7 thereof; FIG. 16 is an enlarged view of portion within the dot-dot-dash broken line box 16 shown in FIG. 15; FIG. 17 is an enlarged cross-sectional view along the line 17-17 in FIG. 5, in the portion within the dot-dot-dash broken line box 17 in the FIG. 7 thereof; FIG. 18 is an enlarged cross-sectional view along the line 18-18 in FIG. 3, in the portion within the dot-dot-dash broken line box 18 in the FIG. 5 thereof, in which the internal mechanism is omitted; FIG. 19 is an enlarged cross-sectional view along the line 19-19 in FIG. 3, in the portion within the dot-dot-dash broken line box 19 in the FIG. 5 thereof, in which the internal mechanism is omitted; FIG. 20 is an enlarged cross-sectional view along the line 20-20 in FIG. 3, in the portion within the dot-dot-dash broken line box 20 in the FIG. S thereof, in which the internal mechanism is omitted; and, FIG. 21 is an enlarged cross-sectional view along the line 21-21 in FIG. 3, in the portion within the dot-dot-dash broken line box 21 in the FIG. 5 thereof, in which the internal mechanism is omitted.

(Continued)



The broken lines in the drawings are for the purpose of illustrating portions of the refrigerator that form no part of the claimed design.

The dot dash broken lines define the bounds of the claimed design and form no part thereof.

The dot-dot-dash broken lines represent the boundaries of the enlarged portion and form no part of the claimed design.

1 Claim, 21 Drawing Sheets

(58) Field of Classification Search

CPC F25D 23/028; F25D 29/00; F25D 17/065;
F25D 23/02; F25D 11/02; F25D 23/126;
F25D 25/025; F25D 23/04; F25D 29/005;
F25D 25/02; F25D 2201/14; F25D 23/12;
F25D 17/045; F25D 2400/04; F25D
2700/12; F25D 2323/021; F25D 2400/06;
F25D 2700/02; F25D 23/062; F25D
11/00; F25D 2400/361; F25D 21/14;
F25D 2323/024; F25D 27/00; F25D
17/042; F25D 2331/803; F25D 23/067;
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3/08; F25D 11/022; F25D 2500/02; F25D
17/062; F25D 21/04; F25D 2400/36;
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2317/067; F25D 23/066; F25D 23/087;
F25D 3/04; F25D 23/00; F25D 2323/023;
F25D 23/006; F25D 23/085; F25D
23/003; F25D 21/08; F25D 31/007; F25D
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F25D 17/047; F25D 11/027; F25D

2400/038; F25D 2317/0661; F25D
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F25D 17/00; F25D 2201/1282; F25D
2303/0842; F25D 2317/04111; F25D
2323/061; F25D 2303/0832; F25D 5/00
See application file for complete search history.

(56)

References Cited

U.S. PATENT DOCUMENTS

D732,090	S	*	6/2015	Sin		D15/89
D786,946	S	*	5/2017	Kim		D15/85
D839,316	S	*	1/2019	Choi		D15/85
D839,324	S	*	1/2019	Choi		D15/91
D880,544	S	*	4/2020	Kim		F25D 25/025 D15/91
D905,767	S	*	12/2020	Lee		D15/89
D921,065	S	*	6/2021	Lee		D15/89

* cited by examiner

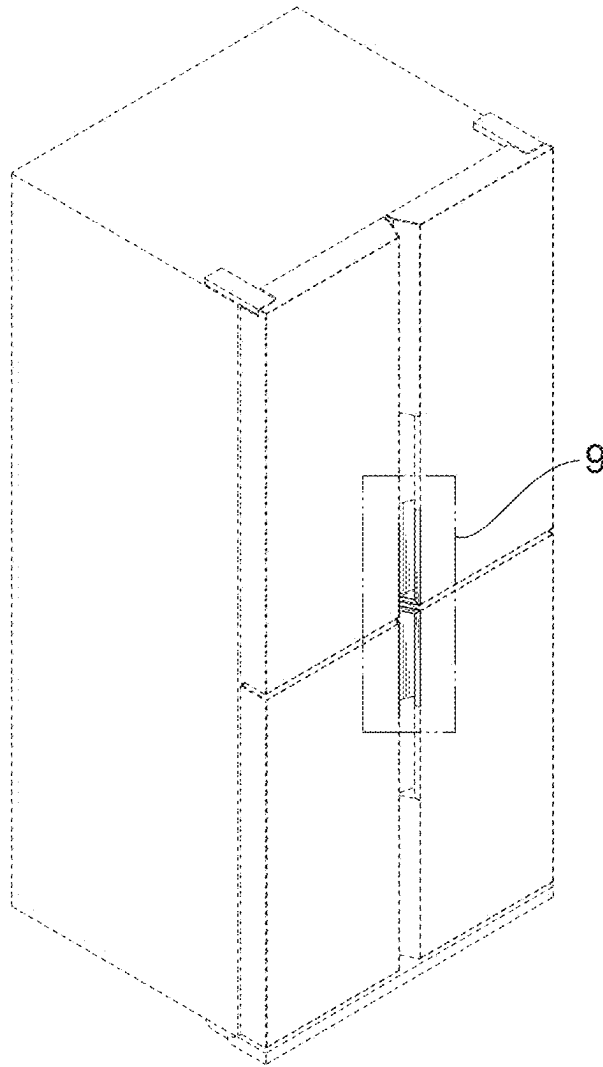


FIG. 1

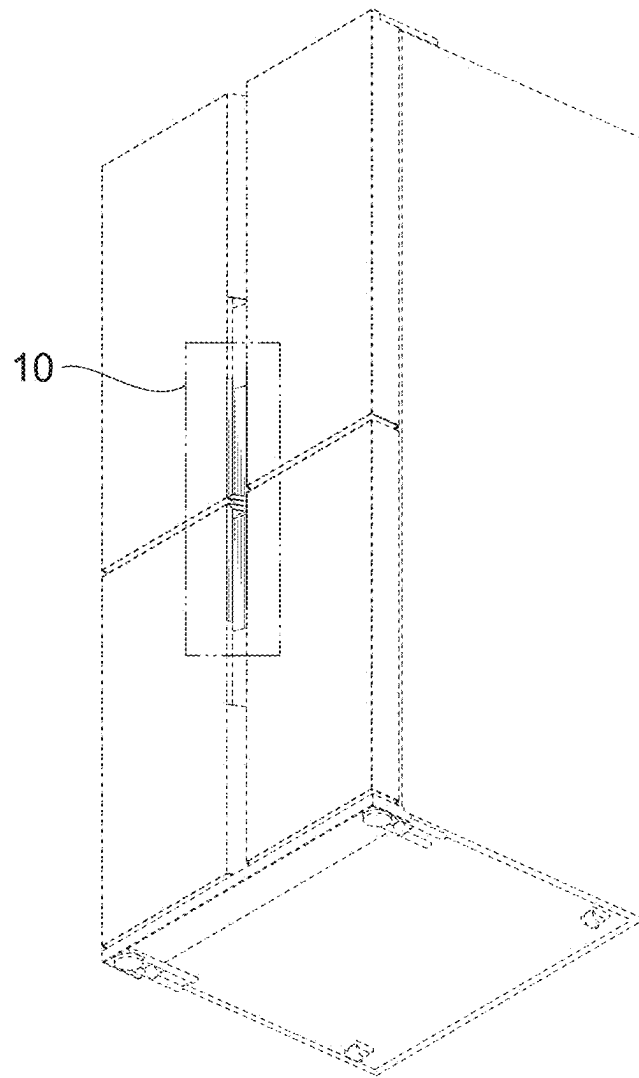


FIG. 2

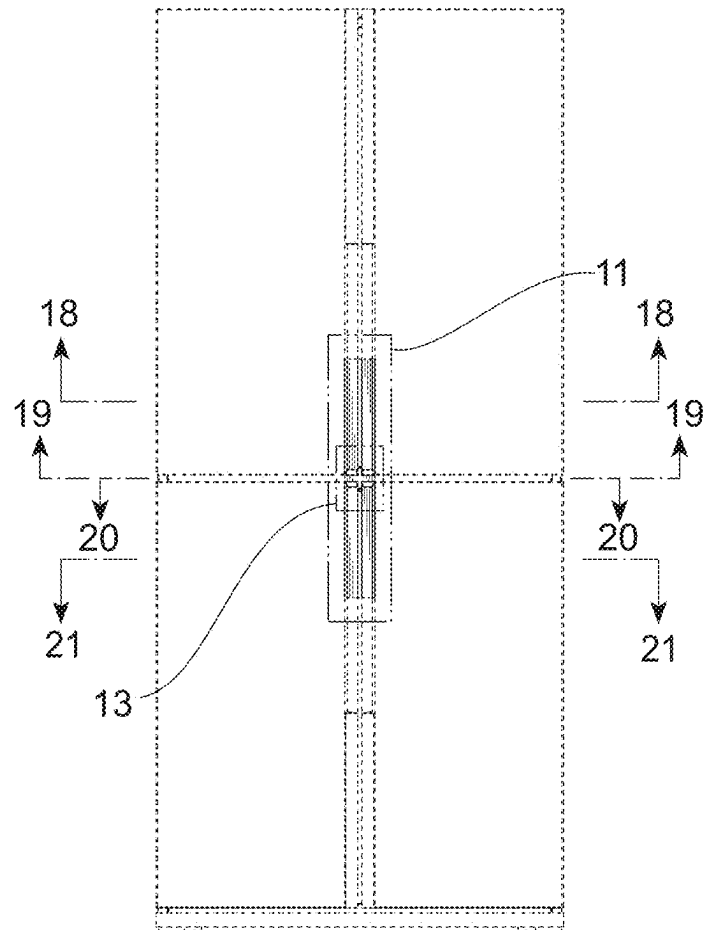


FIG. 3

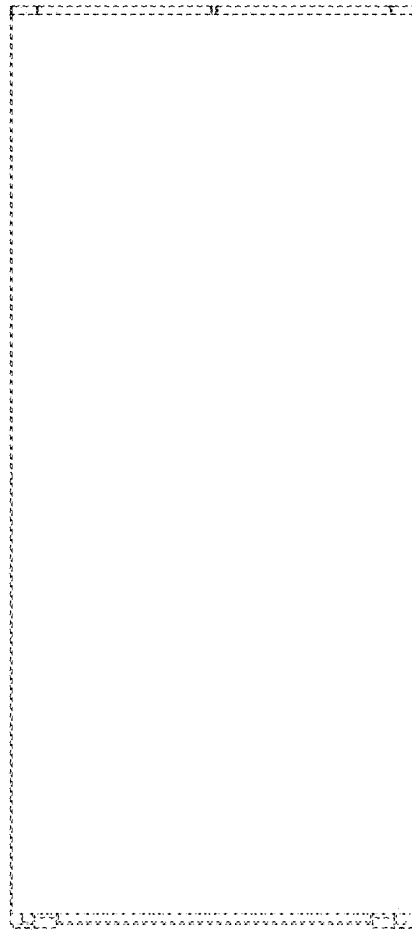


FIG. 4

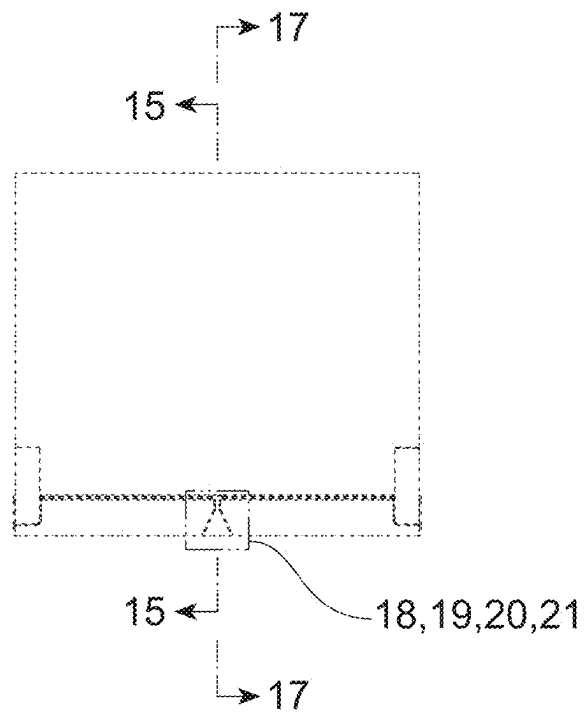


FIG. 5

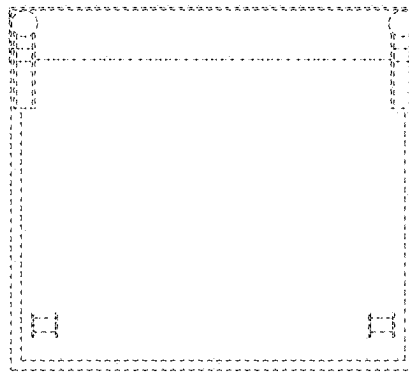


FIG. 6

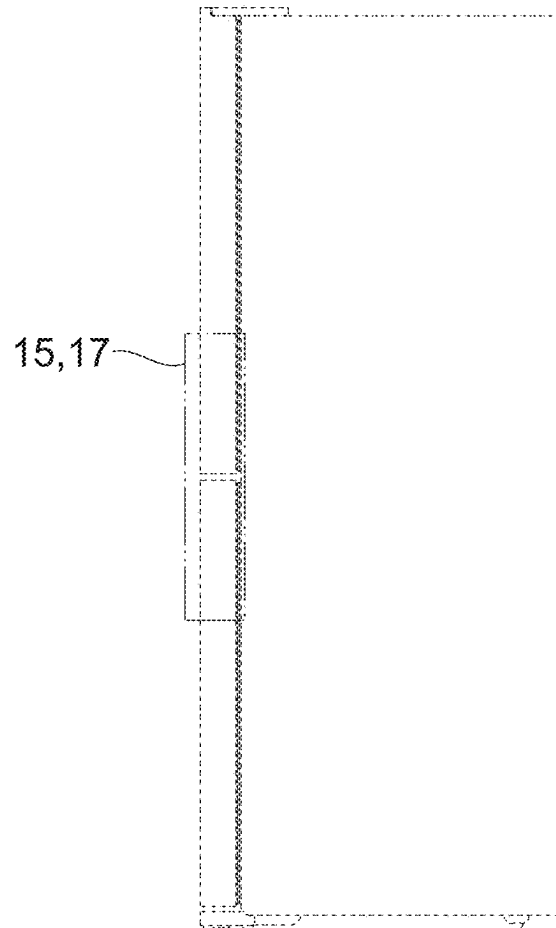


FIG. 7

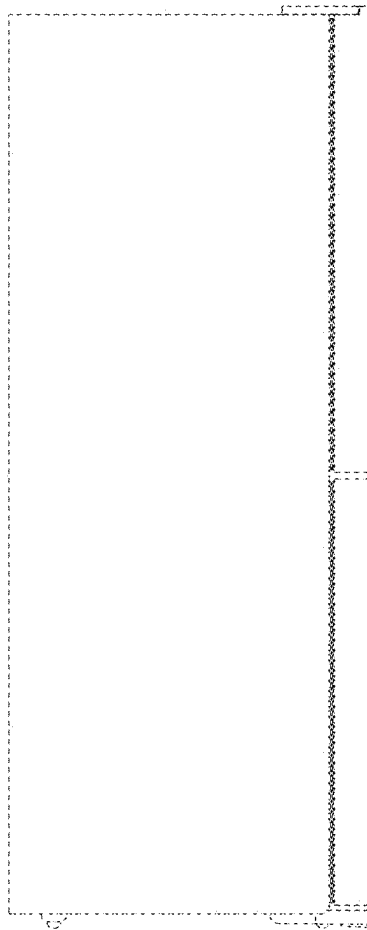


FIG. 8

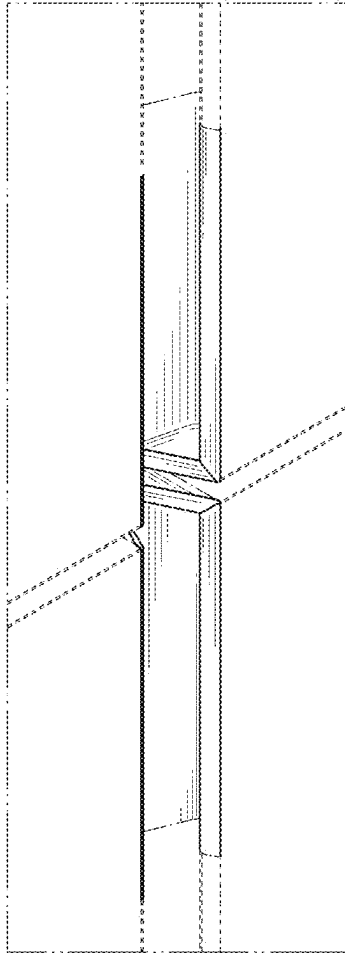


FIG. 9

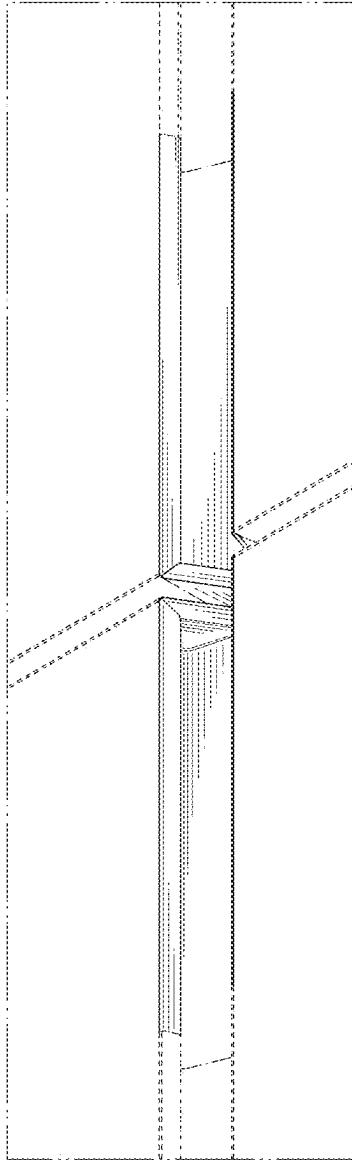


FIG. 10

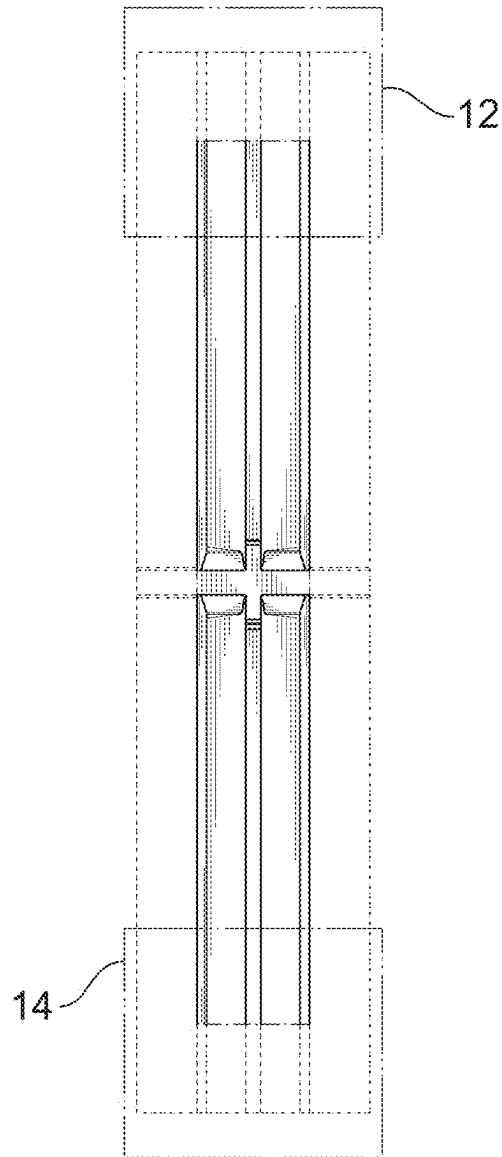


FIG. 11

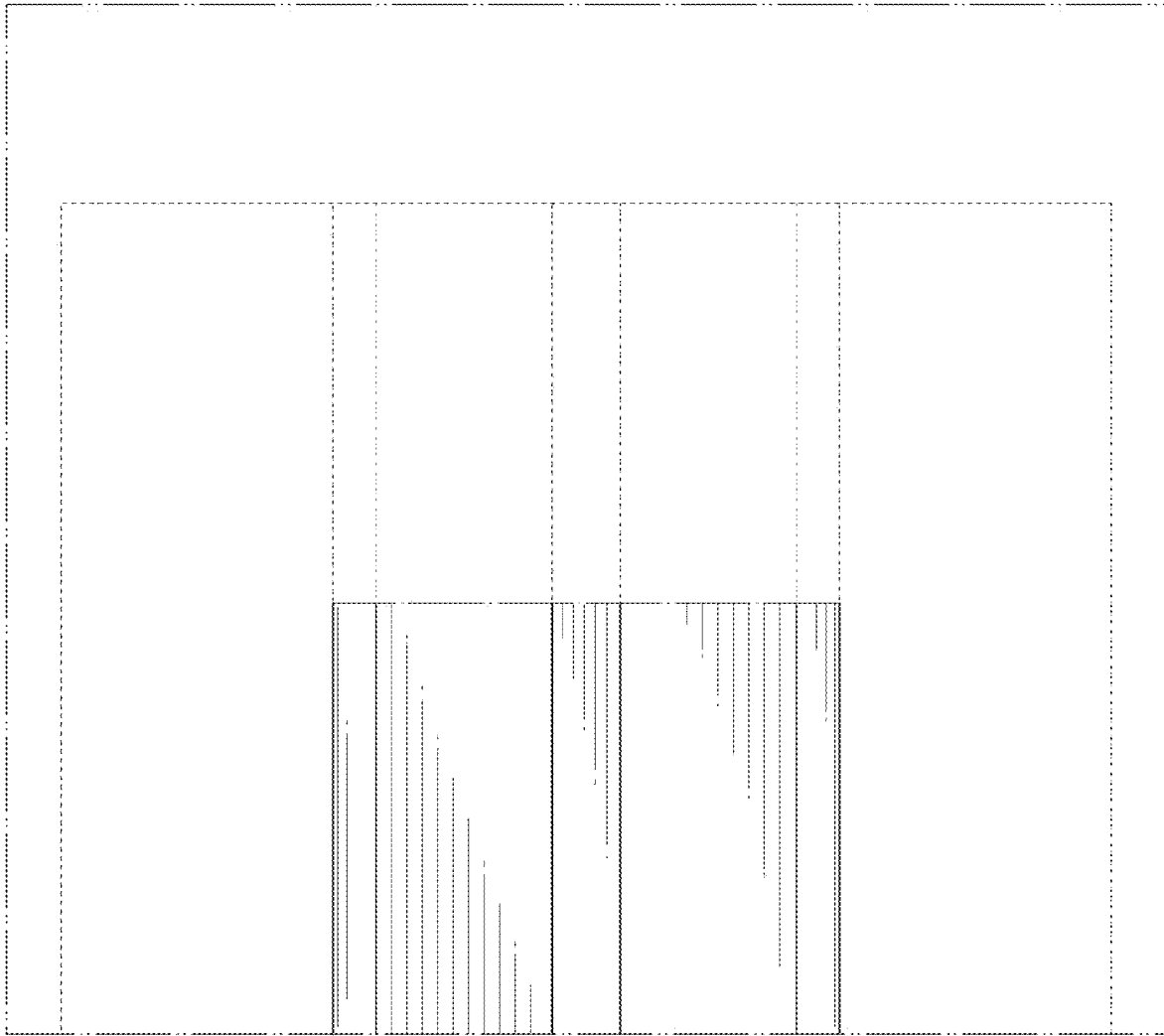


FIG. 12

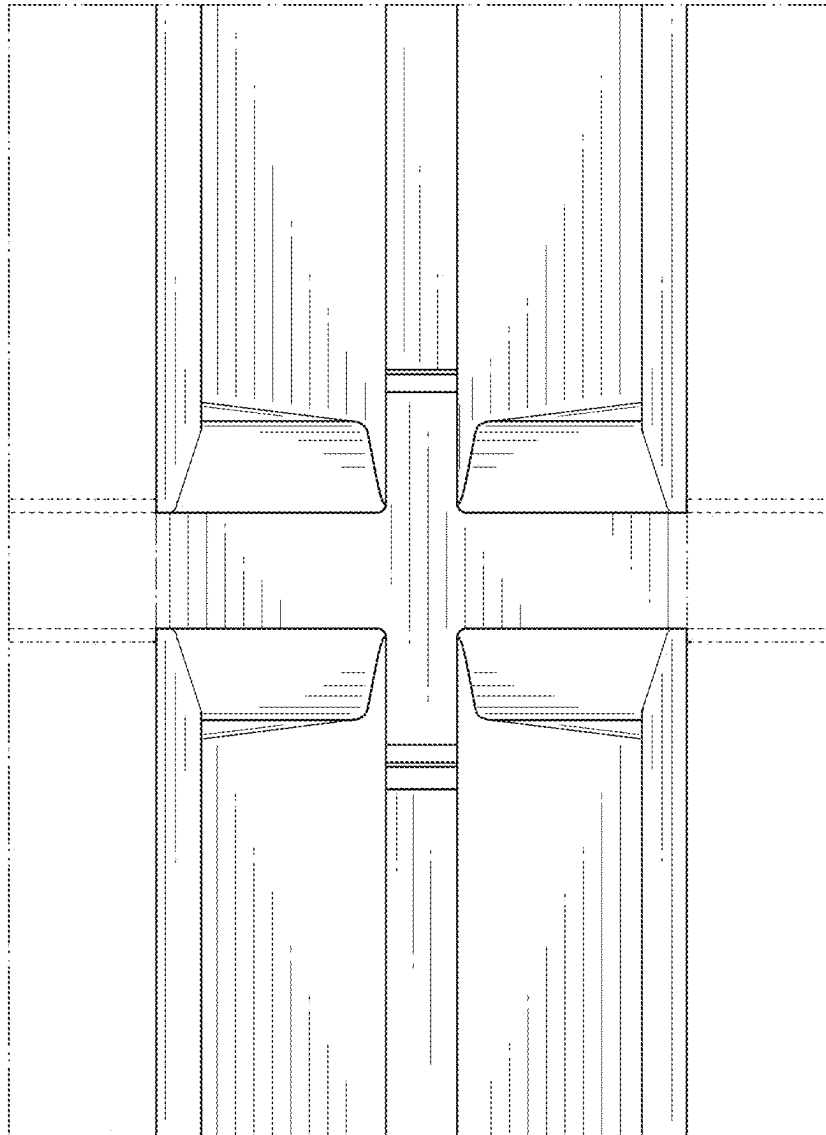


FIG. 13

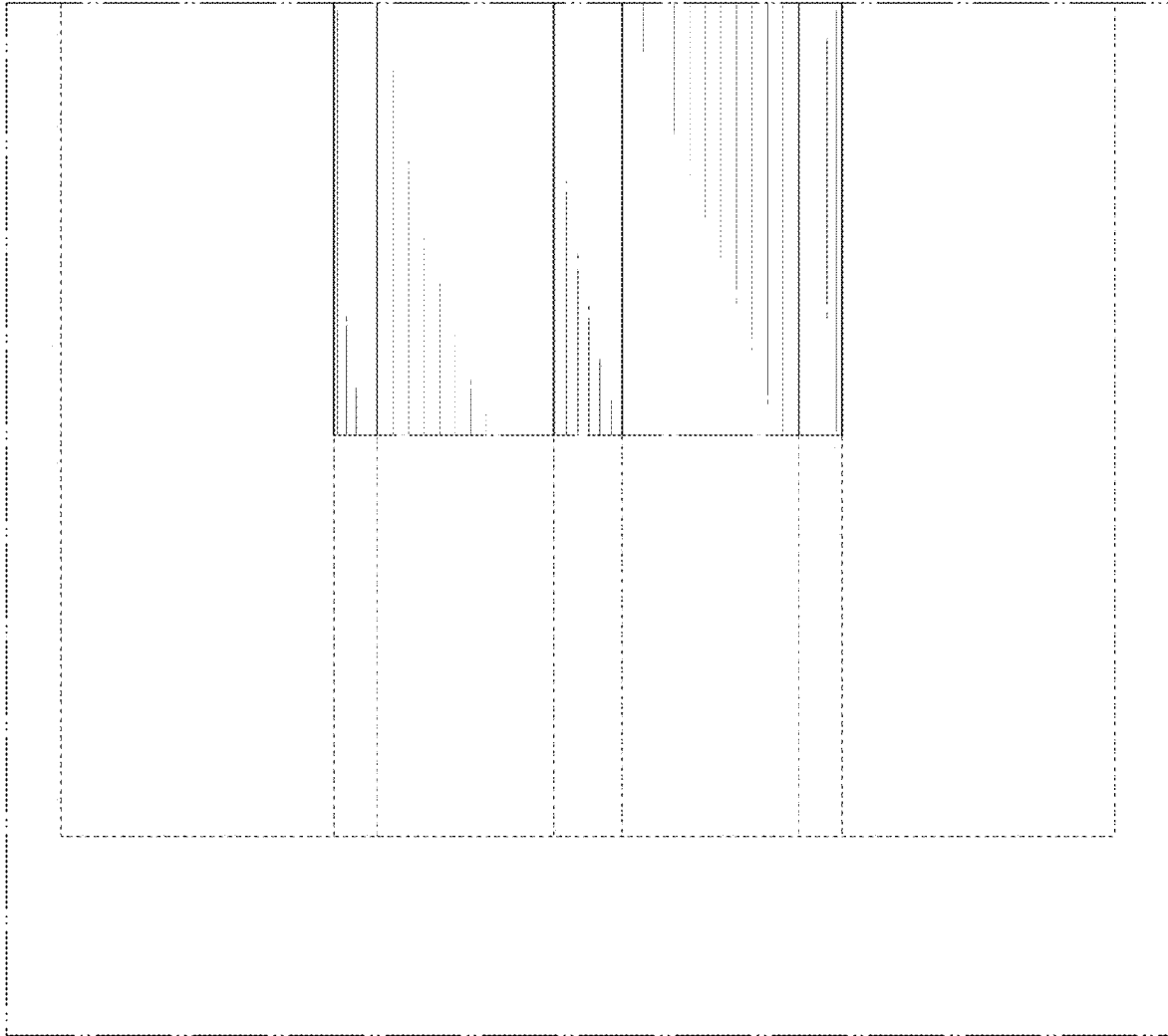


FIG. 14

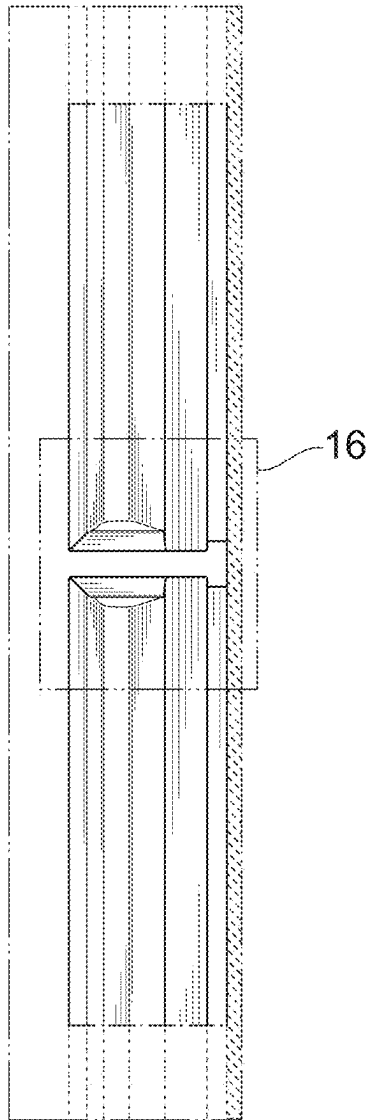


FIG. 15

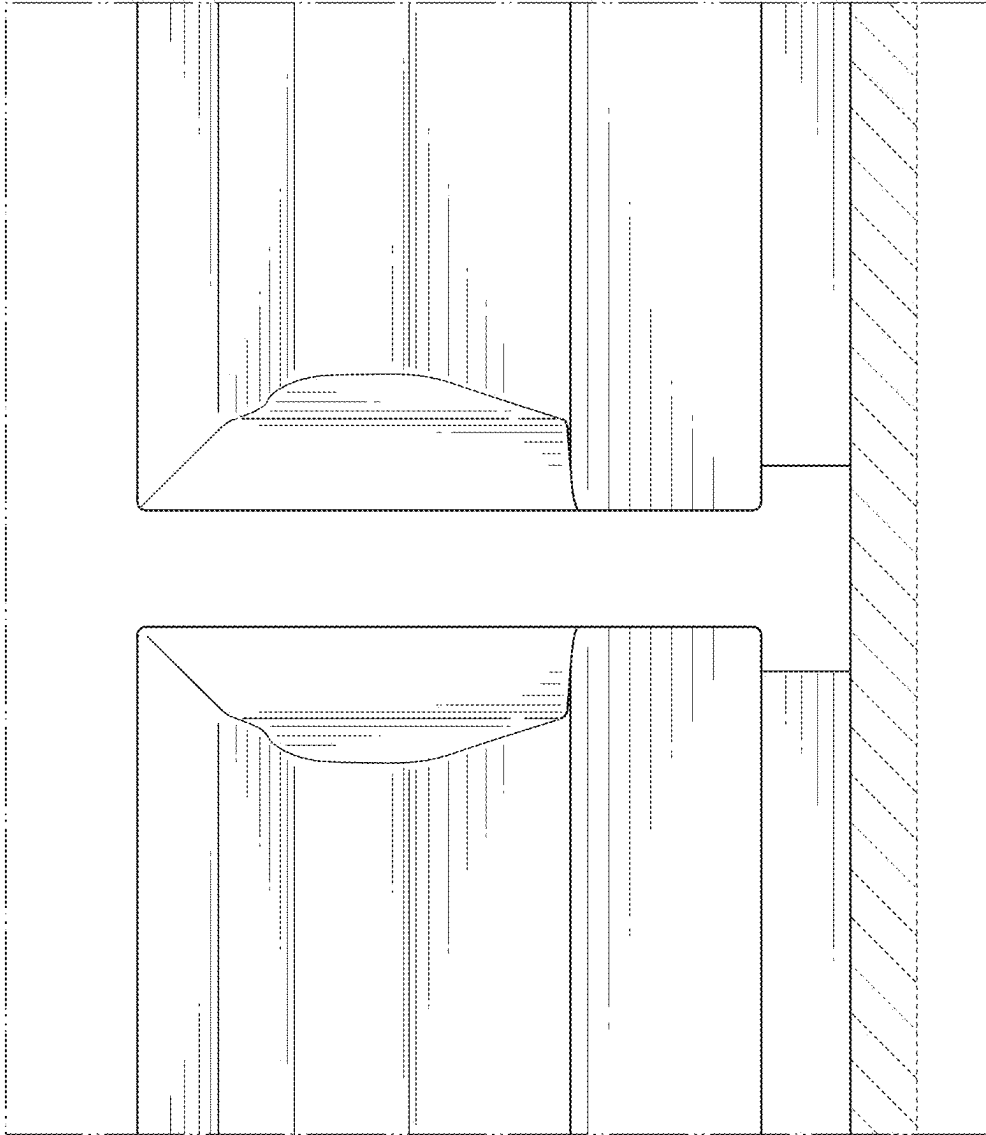


FIG. 16

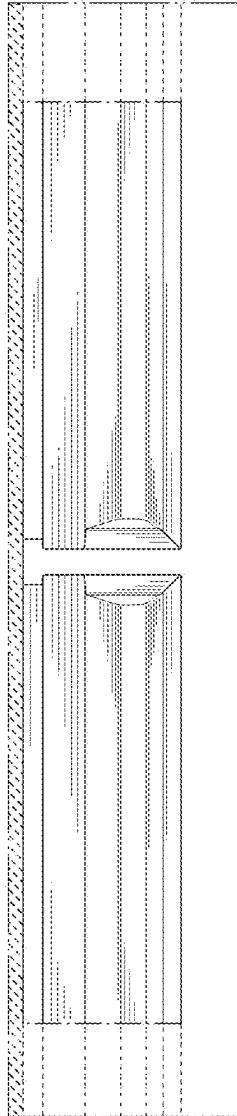


FIG. 17

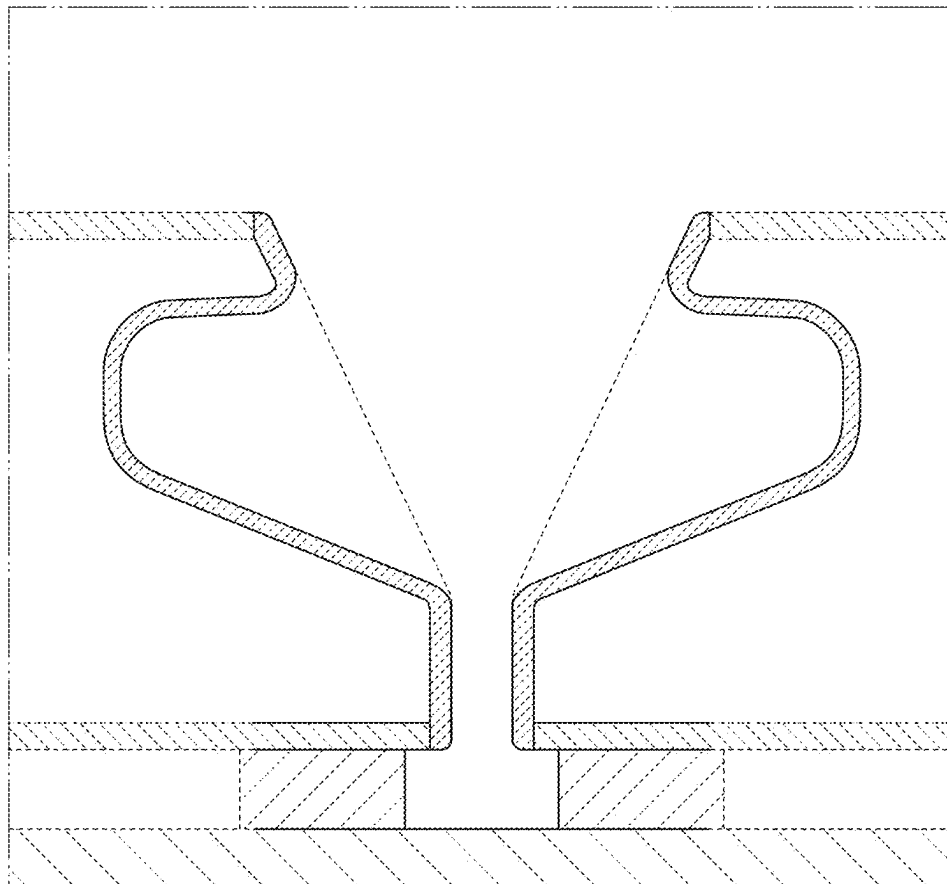


FIG. 18

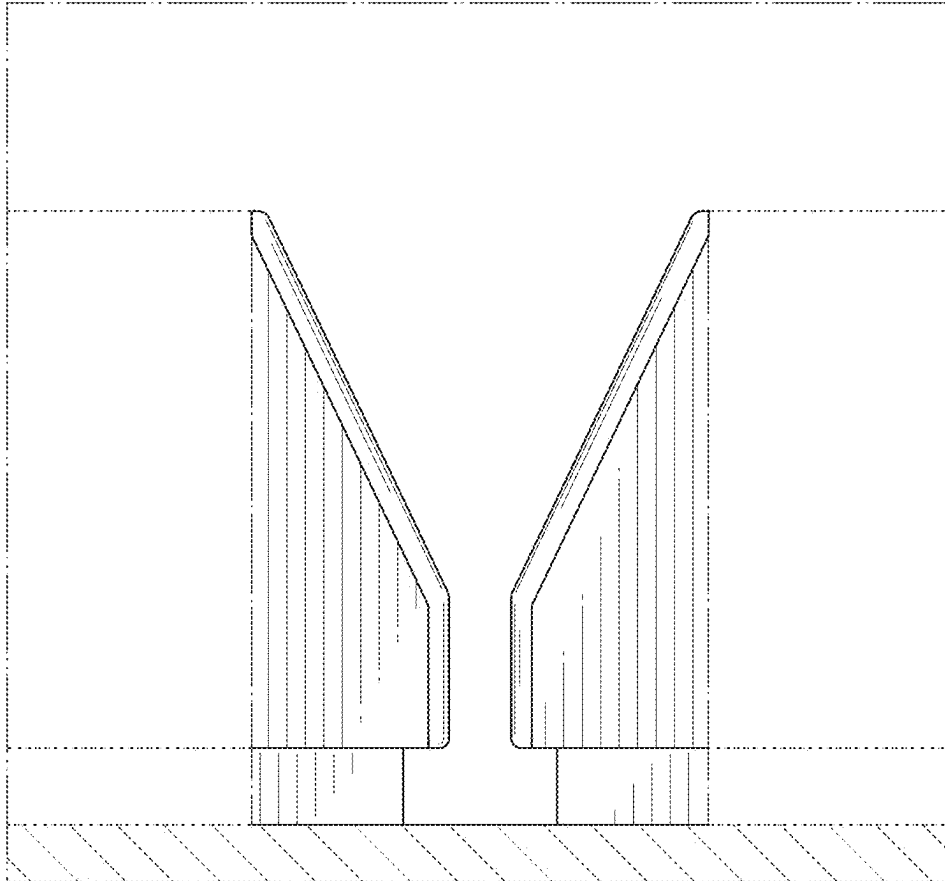


FIG. 19

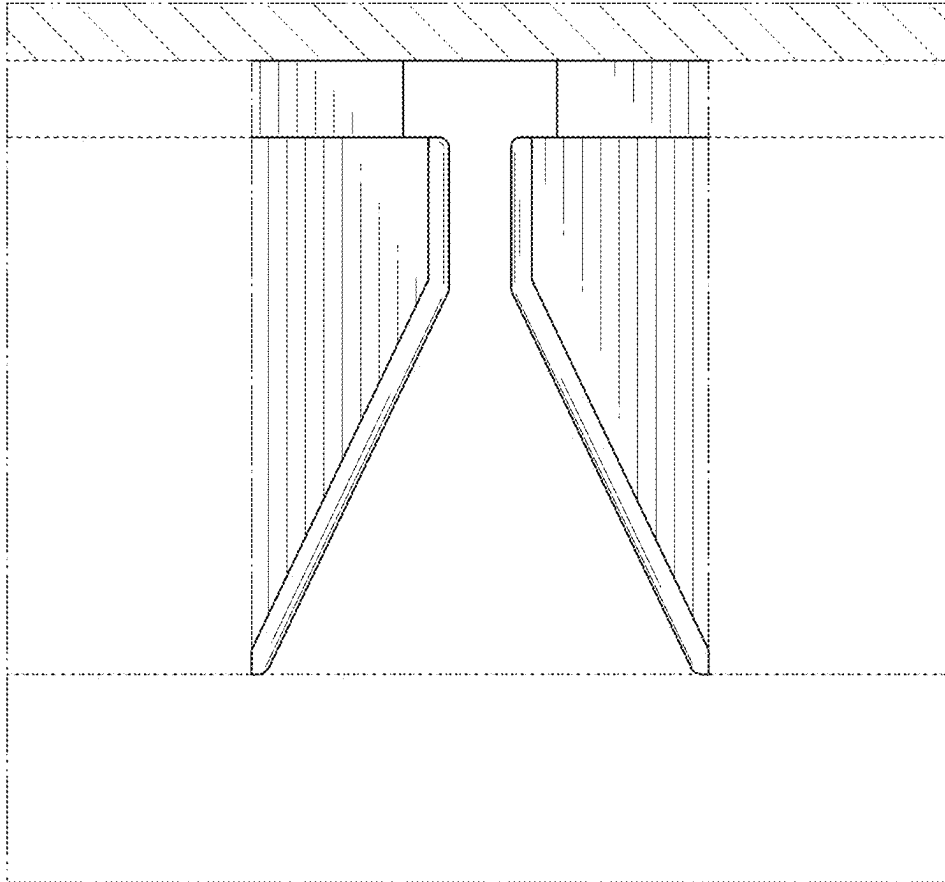


FIG. 20

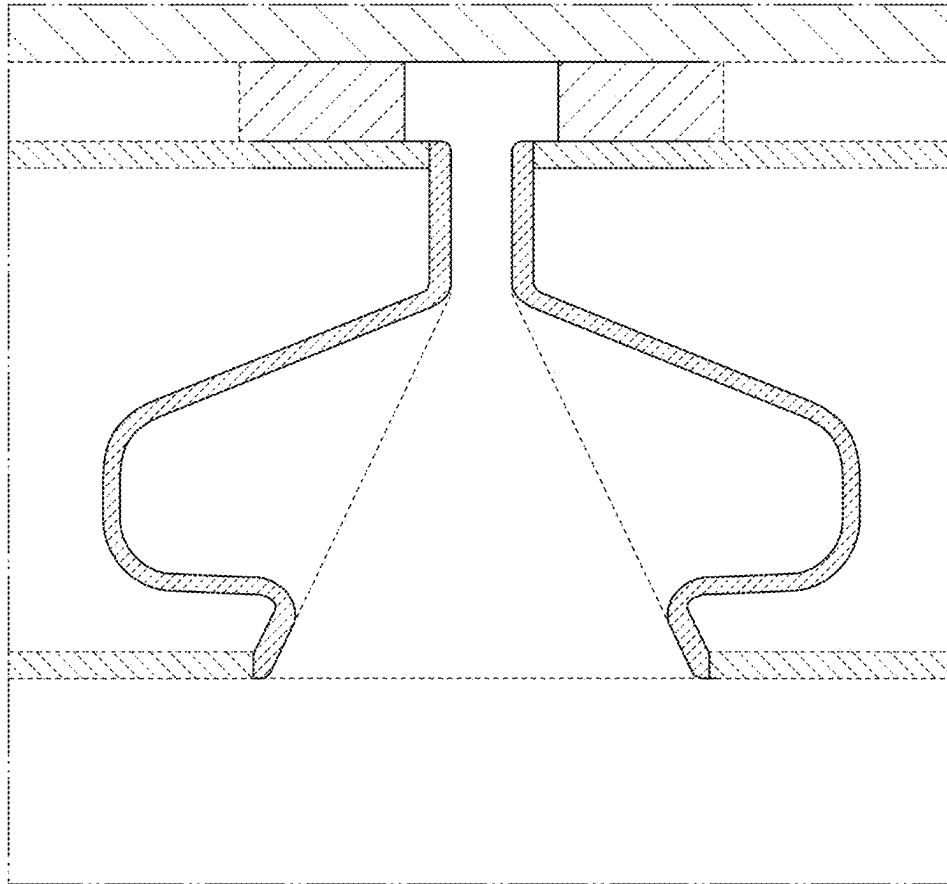


FIG. 21